

ENTERPRISE CAPSTONE PROJECT PROGRAM

UST Global

AI Agents · Multi-Domain · Enterprise Workflows

TELECOM

HEALTHCARE

FINANCE

Bridging AI Innovation with Real-World Enterprise Problem Solving

Document Type	Program Reference Guide
Audience	Learners, Mentors & Evaluators
Program	Enterprise Capstone — AI Agents & Workflows
Domains	Telecom · Healthcare · Finance

1. Program Overview

The UST Enterprise Capstone Project Program challenges learners to design, build, and present production-grade AI agent systems across three core enterprise domains: Telecom, Healthcare, and Finance. Each team selects a domain problem and constructs a fully orchestrated multi-agent workflow, incorporating Retrieval-Augmented Generation (RAG), human-in-the-loop checkpoints, tool usage, and robust error handling.

Projects are evaluated on technical architecture, responsible AI design, workflow sophistication, system robustness, and the quality of live demonstrations and team presentations.

2. Common Technical Expectations

Every project — regardless of domain — must demonstrate the following capabilities:

Capability	Description
Multi-Agent Workflows	Design and implement coordinated agents with clearly defined roles and responsibilities.
Sequential + Parallel Execution	Demonstrate both linear pipelines and concurrent agent task execution where appropriate.
Router-Based Decision Making	Use conditional routing logic to direct data or tasks between agents dynamically.
RAG Pipelines	Integrate document retrieval with LLM generation using vector databases and embedding models.
Human Approval Step	Include at least one human-in-the-loop checkpoint for critical decisions or outputs.
Tool-Using Agents	Agents must invoke external tools (APIs, search, calculators, databases, etc.).
Memory / Context Handling	Maintain relevant conversational or task state across agent interactions.
Error Handling and Retries	Implement graceful failure modes with retry logic and fallback strategies.
Basic Logging and Monitoring	Log agent decisions and actions; surface key metrics for observability.

Recommended Tech Stack

Layer	Suggested Tools
Agent Framework	LangGraph / CrewAI + LangChain
LLM	OpenAI / Gemini
RAG	LangChain / LlamaIndex / LangGraph / CrewAI
Vector DB	ChromaDB / FAISS

TELECOM DOMAIN PROJECTS

only 1/5
5 Projects · Network, Customer & Fraud Intelligence

1. AI-Based Telecom Incident Ticket Analyzer

Problem: Telecom teams receive hundreds of issue tickets daily and struggle to prioritize and resolve them efficiently.

Synthetic Data Learners Can Create:

- Synthetic ticket dataset
- Synthetic outage logs
- Sample telecom PDFs

• Ticket Classification Agent • Severity Detection Agent • Resolution Suggestion Agent • Human Approval Agent	• Sequential • Router-based decision flow	Telecom troubleshooting PDFs	Ticket summarization • Priority classification • Suggested resolution generation

2. Telecom Customer Complaint Resolution Assistant

Problem: Customer complaints are high in volume and difficult to analyze or respond to consistently without manual effort.

Synthetic Data Learners Can Create:

- Synthetic complaint chats
- Complaint history dataset

• Sentiment Agent • Complaint Categorization Agent • Response Generation Agent	• Parallel execution • Sequential orchestration	Complaint policy documents	Complaint summarization • Sentiment analysis • Suggested response generation

3. Telecom Network Outage RCA Assistant

Problem: Engineers spend significant time identifying root causes during network outages, delaying recovery.

Synthetic Data Learners Can Create:

- Synthetic network logs
- Sample outage reports

• Log Analysis Agent • RCA Agent • Recommendation Agent	• Parallel log analysis • Router for escalation	Network incident knowledge base	Root-cause explanation • Incident summary • Suggested troubleshooting steps

4. Telecom Fraud Alert Investigation Assistant

Problem: Fraud alerts require time-consuming manual investigation by analysts, increasing exposure time.

Synthetic Data Learners Can Create:

- Fake call records
- Synthetic fraud cases

• Fraud Detection Agent • Risk Analysis Agent • Report Generation Agent	• Sequential workflow • Human checkpoint	Fraud investigation guidelines	• Fraud summarization • Risk scoring • Investigation report generation

5. AI-Powered Telecom Knowledge Assistant

Problem: Engineers struggle to search and extract information from lengthy telecom manuals and SOPs.

Synthetic Data Learners Can Create:

- Telecom SOP PDFs
- Device manuals

• Retrieval Agent • Answer Validation Agent • Summarization Agent	• RAG pipeline • Memory-aware chatbot	Telecom manuals & SOP documents	• AI chatbot • Context-aware answers • Document summarization

HEALTHCARE DOMAIN PROJECTS

only 1/5
5 Projects · Clinical, Research & Operations Intelligence

1. AI Clinical Report Summarization Assistant

Problem: Doctors spend excessive time reading long clinical reports, delaying patient care decisions.

Synthetic Data Learners Can Create:

- Synthetic patient reports
- Lab test PDFs

• Report Analysis Agent • Summary Agent • Recommendation Agent	• Sequential workflow	Medical guideline PDFs	• Report summarization • Highlight abnormal findings • Follow-up recommendations

2. AI Medical Research Paper Assistant

Problem: Researchers struggle to efficiently analyze and extract insights from large volumes of medical papers.

Synthetic Data Learners Can Create:

- Synthetic research PDFs
- Medical article dataset

• Paper Reader Agent • Insight Agent • Citation Agent	• Parallel extraction • RAG pipeline	Medical research corpus	• Research summarization • Key findings extraction • Q&A chatbot

3. Hospital Resource Allocation Assistant

Problem: Hospitals struggle to allocate beds and staff effectively, especially during peak demand or emergencies.

Synthetic Data Learners Can Create:

- Synthetic hospital datasets

• Bed Allocation Agent • Staff Allocation Agent • Emergency Routing Agent	• Router-based workflow	Hospital policy documents	• Resource allocation suggestions • Emergency prioritization

4. AI Patient Follow-Up Assistant

Problem: Hospitals frequently miss timely follow-up communication with discharged patients.

Synthetic Data Learners Can Create:

- Synthetic patient follow-up data

• Patient Summary Agent • Reminder Agent • Escalation Agent	• Sequential workflow • Memory-based context	Care plan & discharge guidelines	• Personalized reminders • Follow-up summaries

5. Healthcare Insurance Claim Review Assistant

Problem: Insurance claims are manually reviewed, creating bottlenecks and high error rates.

Synthetic Data Learners Can Create:

- Synthetic claims PDFs
- Fake insurance forms

• OCR Agent • Validation Agent • Recommendation Agent	• Sequential + human approval	Insurance policy guidelines	• Claim summarization • Missing field detection • Approval recommendation

FINANCE DOMAIN PROJECTS

only 1/5
5 Projects · Risk, Compliance & Investment Intelligence

1. AI Loan Eligibility Assessment Assistant

Problem: Banks manually review loan applications, slowing approvals and introducing inconsistency.

Synthetic Data Learners Can Create:

- Synthetic customer financial data
- Loan application PDFs

• Financial Analysis Agent • Risk Agent • Recommendation Agent	• Sequential • Router-based workflow	Lending policy documents	• Loan summarization • Risk explanation • Eligibility recommendation

2. AI Financial Document Q&A; Assistant

Problem: Users and analysts struggle to understand dense financial reports and annual statements.

Synthetic Data Learners Can Create:

- Annual reports
- Fake balance sheets

• Retrieval Agent • KPI Extraction Agent • Summary Agent	• RAG workflow • Memory-based context	Financial reports & filings	• RAG chatbot • Financial summarization • KPI extraction

3. AI Fraud Transaction Investigation Assistant

Problem: Fraud analysts manually inspect suspicious transactions, making real-time detection difficult.

Synthetic Data Learners Can Create:

- Synthetic transaction dataset

• Transaction Agent • Fraud Detection Agent • Reporting Agent	• Parallel fraud analysis	Fraud detection rulebooks	• Fraud explanation • Risk scoring • Investigation reports

4. AI Expense Audit Assistant

Problem: Expense auditing requires significant manual effort, prone to missed violations and duplicates.

Synthetic Data Learners Can Create:

- Synthetic invoices
- Fake reimbursement forms

• OCR Agent • Validation Agent • Audit Agent	• Sequential workflow	Expense & reimbursement policies	• Expense summarization • Duplicate detection • Policy violation detection

5. AI Investment Research Assistant

Problem: Analysts spend excessive time reviewing multiple market reports to synthesize investment insights.

Synthetic Data Learners Can Create:

- Synthetic market reports
- Financial news PDFs

• Market Research Agent • Trend Analysis Agent • Summary Agent	• Parallel research workflow	Market research & financial news	• Market trend summarization • Investment insights • Comparative analysis

6. Final Evaluation Rubric

Projects are scored out of 100 marks across five evaluation dimensions. Teams are expected to demonstrate depth across all categories during the live demo and Q&A; session.

Category	Marks
✓ Technical Implementation & Architecture	25
✓ Use of Generative AI + Agents	20
✓ Workflow Design (Sequential / Parallel / Router)	15
✓ Robustness & Error Handling	15
✓ Presentation, Demo & Team Participation	25
TOTAL	100 ✓

Detailed Scoring Breakdown

Technical Implementation & Architecture — 25 Marks

Criteria	Marks
Architecture clarity and documentation	5
Working implementation (functional demo)	10
RAG quality and relevance	5
Overall system design coherence / pipeline	5

Generative AI + Agent Usage — 20 Marks

Criteria	Marks
Proper use of agents and role separation	5
Prompt engineering quality	5
Memory and context handling	5
* Human-in-the-loop implementation → Responsible use of AI/MAS etc.	5 ✓

Workflow Design — 15 Marks

Criteria	Marks
Sequential workflow implementation	5
Parallel / router workflow implementation	5
Tool usage and orchestration quality	5

Robustness & Error Handling — 15 Marks

Criteria	Marks
* Error handling and graceful failures (fallback)	5
* Logging and monitoring implementation (LangSmith)	5
Workflow reliability and stability	5

Presentation, Demo & Team Participation — 25 Marks

Criteria	Marks
Demo quality and flow - story: PB ... Sol?	5
Architecture explanation clarity	5
Troubleshooting discussion depth	5
Q&A handling and knowledge depth	5
✓ Equal participation from all team members	5

7. Recommended Demo Flow

Each team should structure their presentation and live demo as follows:

#	Section	Duration
1	Problem Statement — Explain the business problem being solved	3 mins
2	Architecture — Walk through system design and agent topology	5 mins
3	Workflow Explanation — Detail sequential, parallel, and router flows	5 mins
4	Live Demo — End-to-end demonstration of the working system	7 mins
5	Challenges Faced — Discuss technical and design challenges	3 mins
6	Q&A — Respond to evaluator and peer questions	7 mins
	Total	30 mins

8. Important Guidance for Learners

1 Create Your Own Synthetic Datasets

Learners are encouraged to generate realistic synthetic data — CSV files, JSON datasets, PDF reports — that simulate enterprise scenarios. This removes dependency on sensitive real-world data.

2 Generate Sample Documents Manually

Create sample PDFs, SOP documents, and knowledge-base files to power your RAG pipelines. The quality of your RAG context directly impacts your system's output quality.

3 Simulate Enterprise Workflows Locally

Focus on demonstrating realistic end-to-end workflows locally. Cloud deployments are not required; a well-designed local system scores higher than a poorly orchestrated cloud deployment.

4 Prioritize Architecture over Scale

Focus on clean, well-documented architecture and working demos over large-scale complexity. Evaluators reward clarity of design and correctness of agent orchestration.

5 Orchestration is the Core Skill

The primary evaluation focus is on how well agents are orchestrated — not on the complexity of integrations. A simple but well-orchestrated multi-agent system is preferred.

This program is designed to develop enterprise-grade AI engineering skills. Approach each project as if you are delivering it to a real client — with clarity, professionalism, and technical rigor.